User Experience

UEX3IL 2

User Research

Intro

UEX3IL 1



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User Research

UEX3IL 2



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Requirements &
Low-Fidelity Prototyping

UEX3IL 3



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Evaluation

UEX3IL 4



Kathrin Probst

High-Fidelity Prototyping

UEX3IL 5



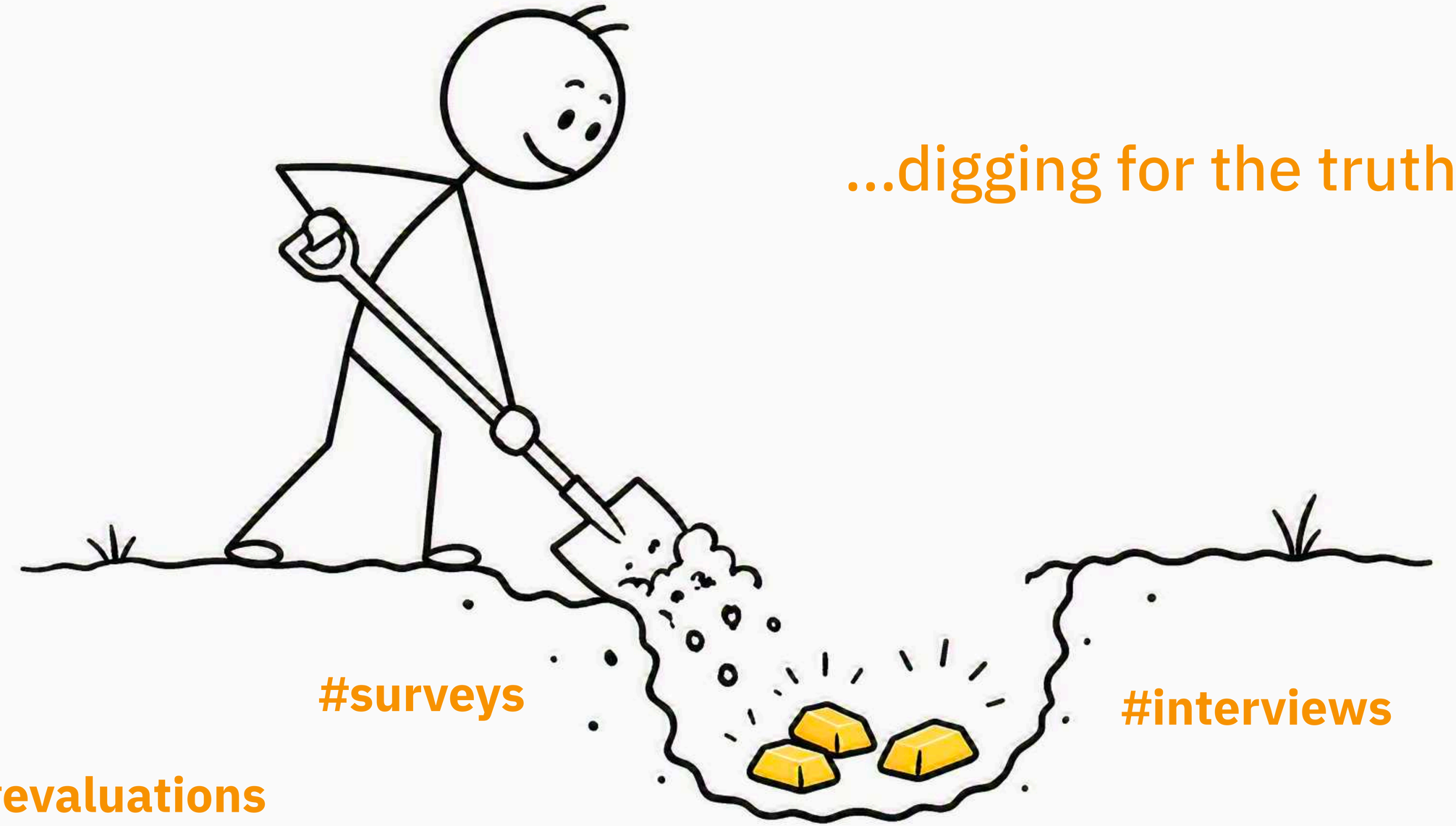
Lisa Freller

Our goal today

UEX3IL 2

- What UX Researcher do?
- What are common User Research methods?
- Question types
- What qualifies as a good interview question? QUIZ
- AI Questionnair + Pilot testing Example
- Affinity Diagramming
- Best Practices (AI)
- Submission

UX Researcher...



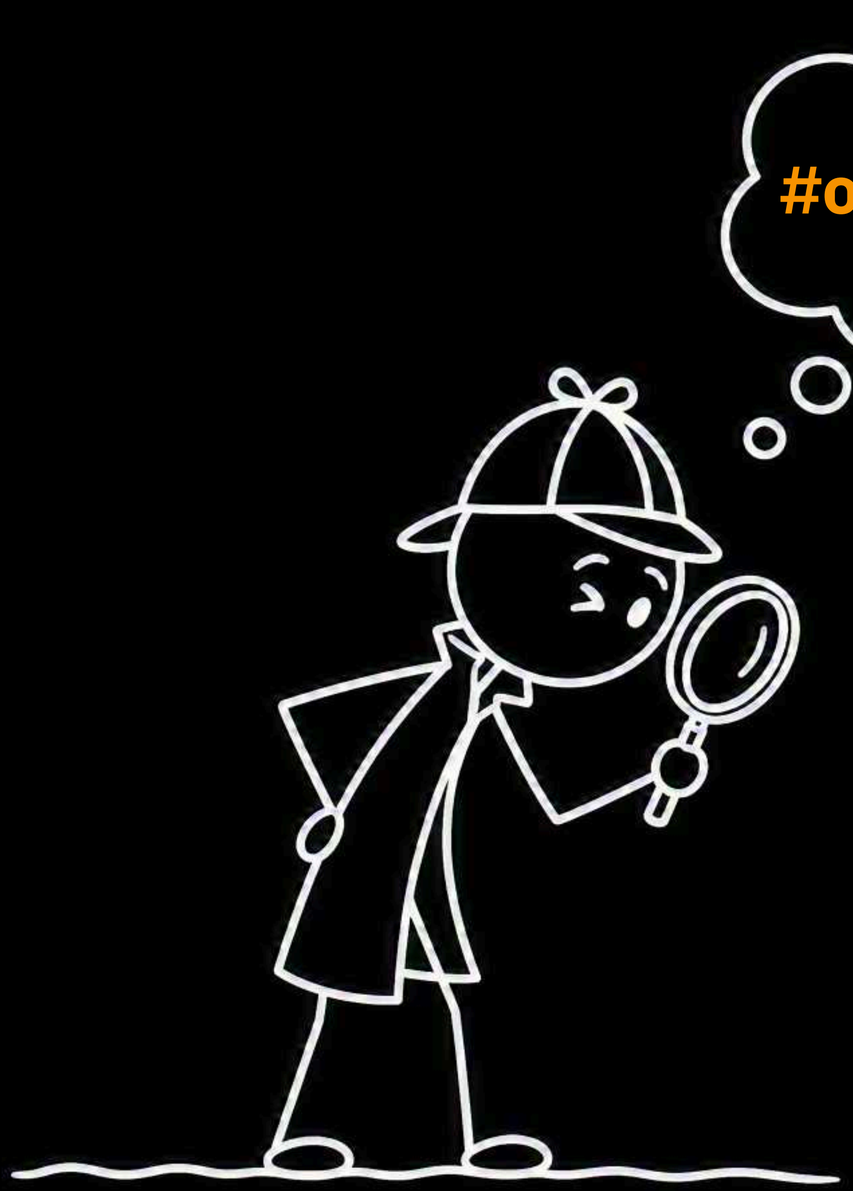
...digging for the truth...

#surveys

#interviews

#evaluations

[pic 1]



...and thinking like a detective.

[pic 2]

Fundamentally, all UX research answers one of two questions:

Who are our users &
What are they trying to do?

Can people use the thing
we have designed
to solve their problem?

[4]p.18

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field visit

usability test

Can people use the thing
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[4]p.18

Fundamentally, all UX research answers one of two questions:

UEX3IL 2 User Research

Who are our users &
What are they trying to do?

field visit

UEX3IL 4 Evaluation

usability test

Can people use the thing
we have designed
to solve their problem?

[4]p.18

User ONLY

How people currently solve their problem?

What are their skills & motivations?



Observe user behaviors

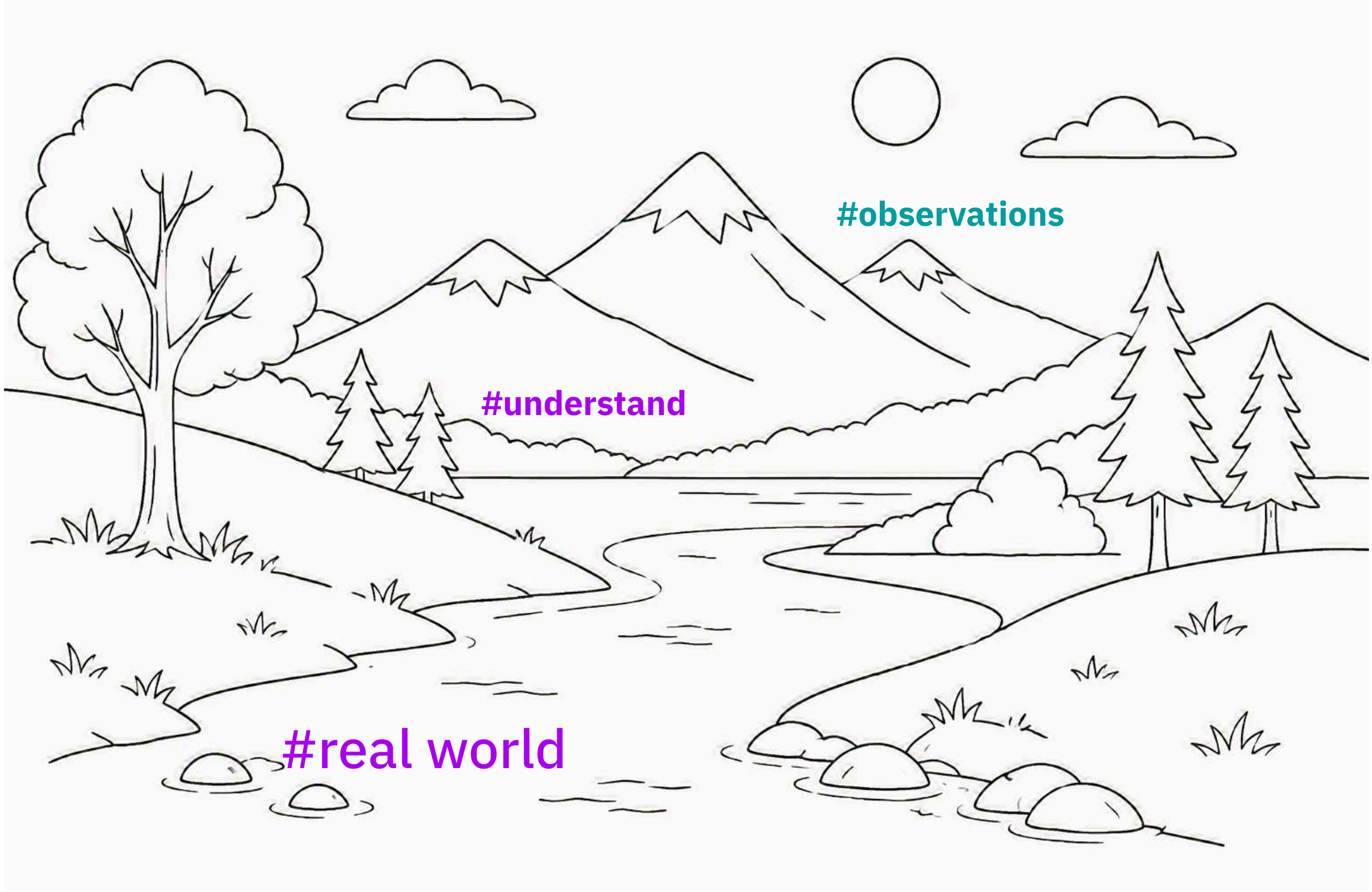
Explore user needs

Find out user goals & pain points

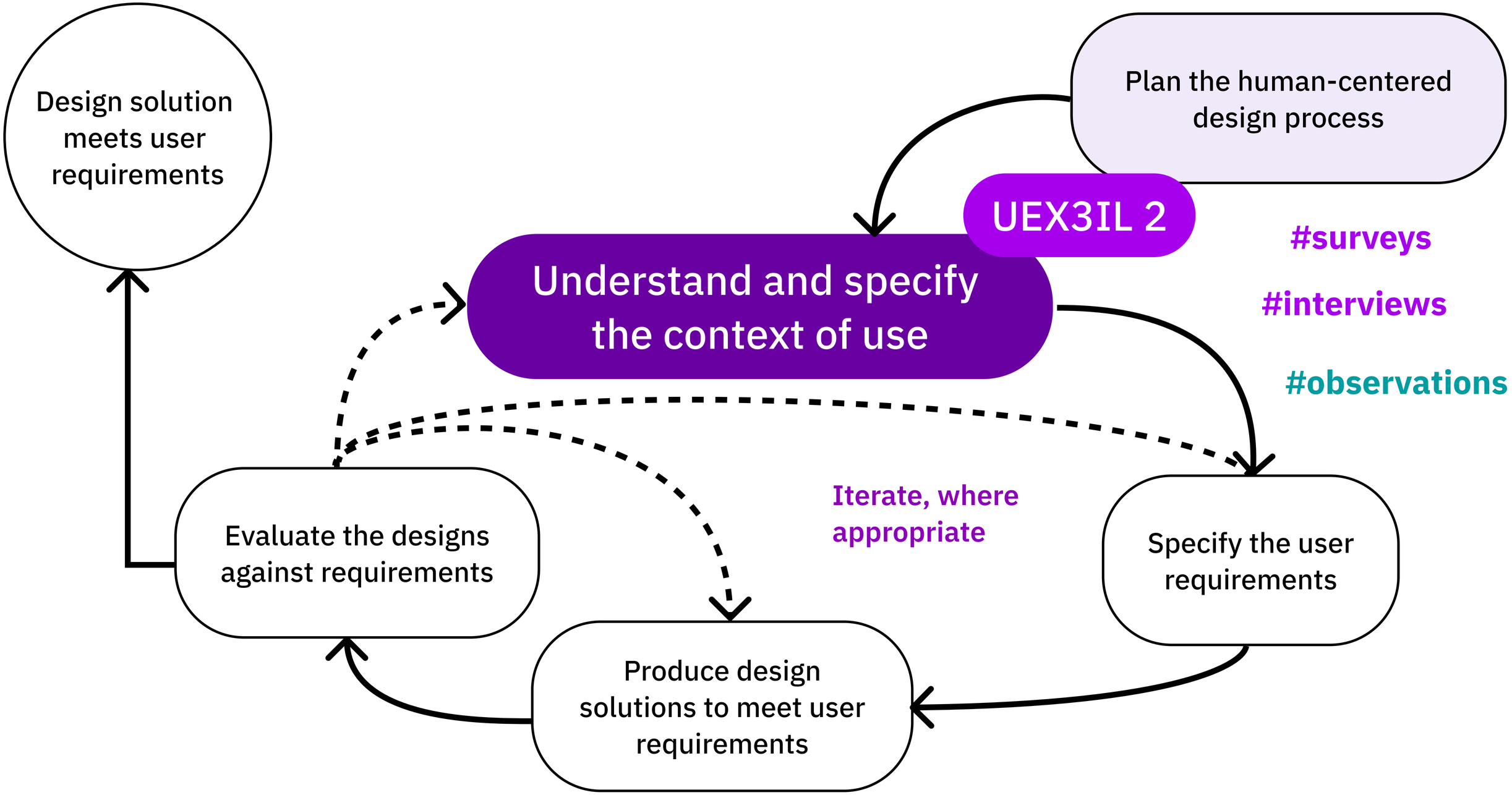
What do they think?

How do they act?

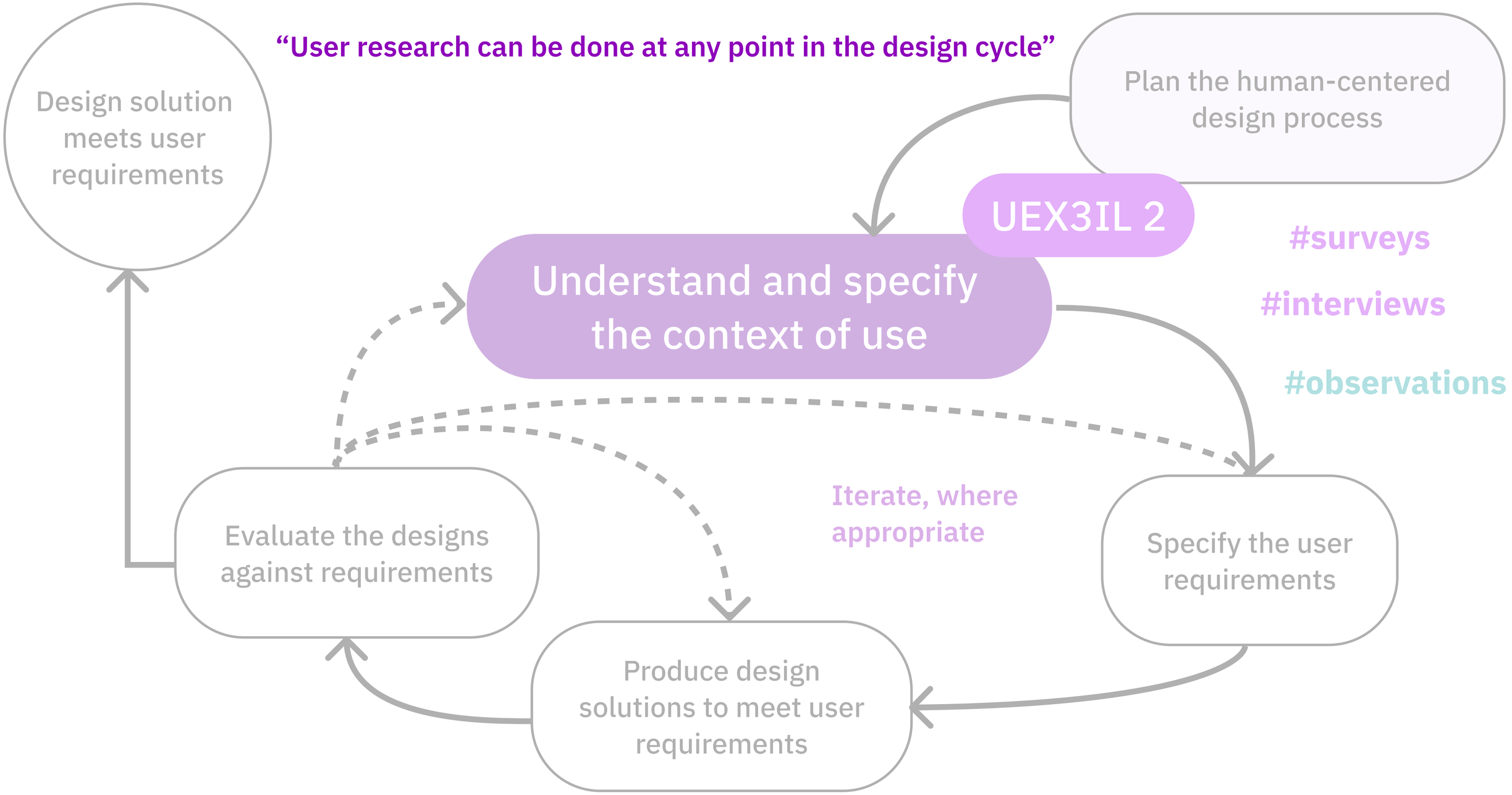
[pic 2]; [4]p.18



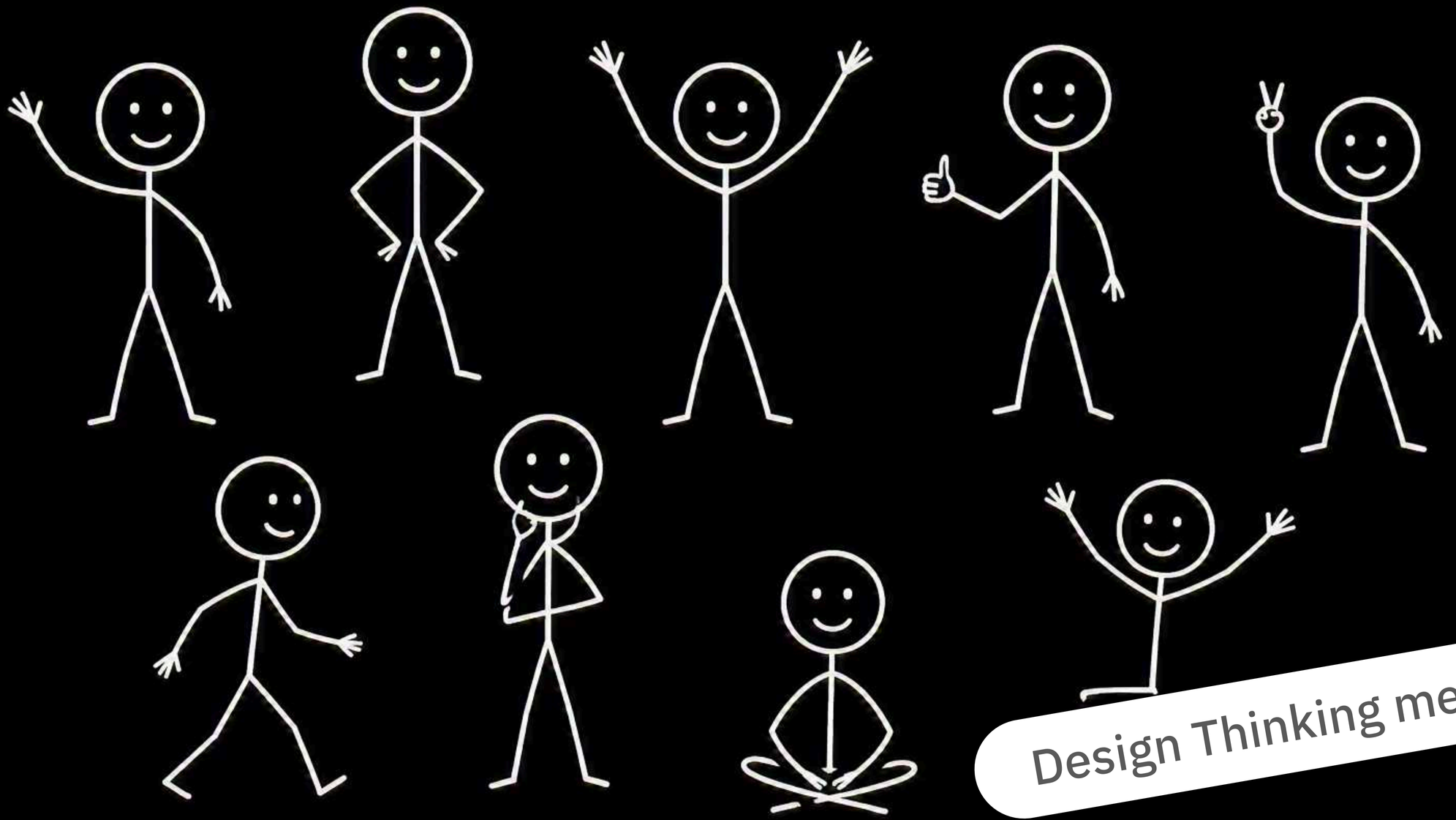
[pic 3]



ISO 9241-11:2018 „Human-Centered Design for Interactive Systems“



ISO 9241-11:2018 „Human-Centered Design for Interactive Systems“



Design Thinking method

[pic 4] [4]p.7

UEX3IL 1

1

Defined the goal

- **Topic**
- **Target audience**
- **Assumptions**
- **Focus areas**

A) Choose user research method(s)

(user interview, survey, contextual inquiry...)

UEX3IL 2

2

A) Choose user research method(s)

(user interview, survey, contextual inquiry...)

B) B1. Use two AI tools + compare them + internal Pilottest

B2. Use one AI tool and iterate your prompts + internal Pilottest

UEX3IL 2

2

A) Choose user research method(s)

(user interview, survey, contextual inquiry...)

B) B1. Use two AI tools + compare them + internal Pilottest

B2. Use one AI tool and iterate your prompts + internal Pilottest

C) Conduct your user research

(2 user interviews of each person in the group)

UEX3IL 2

2

A) Choose user research method(s)

(user interview, survey, contextual inquiry...)

B) B1. Use two AI tools + compare them + internal Pilottest

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C) Conduct your user research

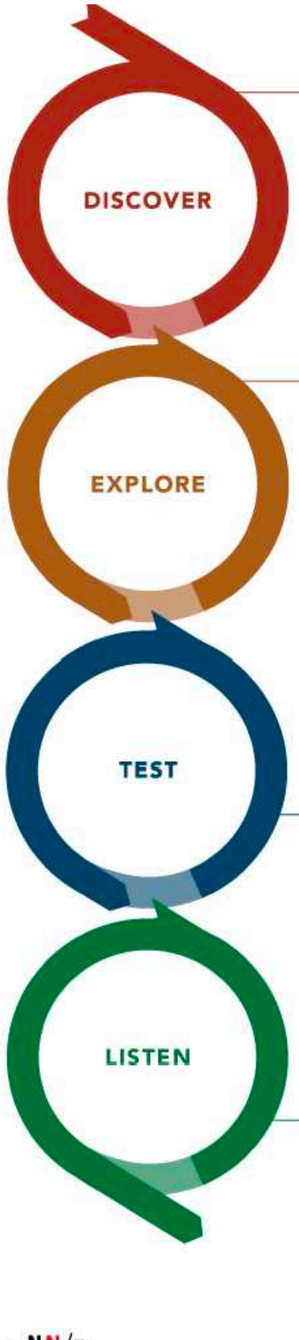
(2 user interviews of each person in the group)

D) Preparation & Analysis

(question feedback, recording, affinity diagramming...)

User research method(s)

UX activities



DISCOVER

METHODS	ACTIVITIES
Field studies/user interviews	Find allies
Diary studies	Talk with experts
Stakeholder interviews	Follow ethical guidelines
Requirements & constraints	Involve stakeholders
Sales & support interviews	Hunt for data sources
Support call monitoring	Determine UX metrics
Competitive testing	

EXPLORE

METHODS	ACTIVITIES
Competitive analysis	Follow Tog's principles of IXD
Design review	Use evidence-based guidelines
Persona building	Design for universal access
Task analysis	Give users control
Journey mapping	Prevent errors
Human-centered design	Improve error messages
Design diversity exploration	Provide helpful defaults
Pluralistic walkthrough	Check for inconsistencies
Prototype feedback & testing	Map features to needs
Write user stories	Make software updating easy
Card sorting	Plan for repair and recycling
	Avoid waste
	Consider diverse contexts
	Look for perverse incentives
	Consider social implications

TEST

METHODS	ACTIVITIES
Qualitative usability testing	Protect personal information
Training research	Keep data safe
User group outreach	Deliver both good and bad news
Social media monitoring	Track usability over time
Forum post analysis	Include diverse users
Benchmark testing	Track usability bugs
Accessibility evaluation	Make training information
Test instructions & help	

LISTEN

METHODS	ACTIVITIES
Surveys	Pay attention to user sentiment
Analytics review	Reduce the need for training
Search-log analysis	Communicate future directions
Usability bug review	Recruit people for future research
Feedback review	
FAQ review	
Conference outreach	
Q&A at talks and demos	

User Research

Evaluation

Bold methods are some of the most commonly used.

NNGROUP.COM NN/g

Generative vs. Evaluative Research

Generative Research
(investigate user needs for new solutions)

“Develop a deeper understanding of your users and the problem you´re trying to solve for them.”[7]

Qualitative user data

- Focus Groups
- User interviews

Evaluative Research
(validates existing designs for improvements)

“Evaluate a product or concept and collect data that helps improve your solution.”[7]

Quantitative & Qualitative user data

- Usability Testing
- A/B Testing

[7]

Attitudinal vs. Behavioral



cf. [8]

Qualitative vs. Quantitative

Why & how to fix

is descriptive and conceptual, helps understand the ‘why’

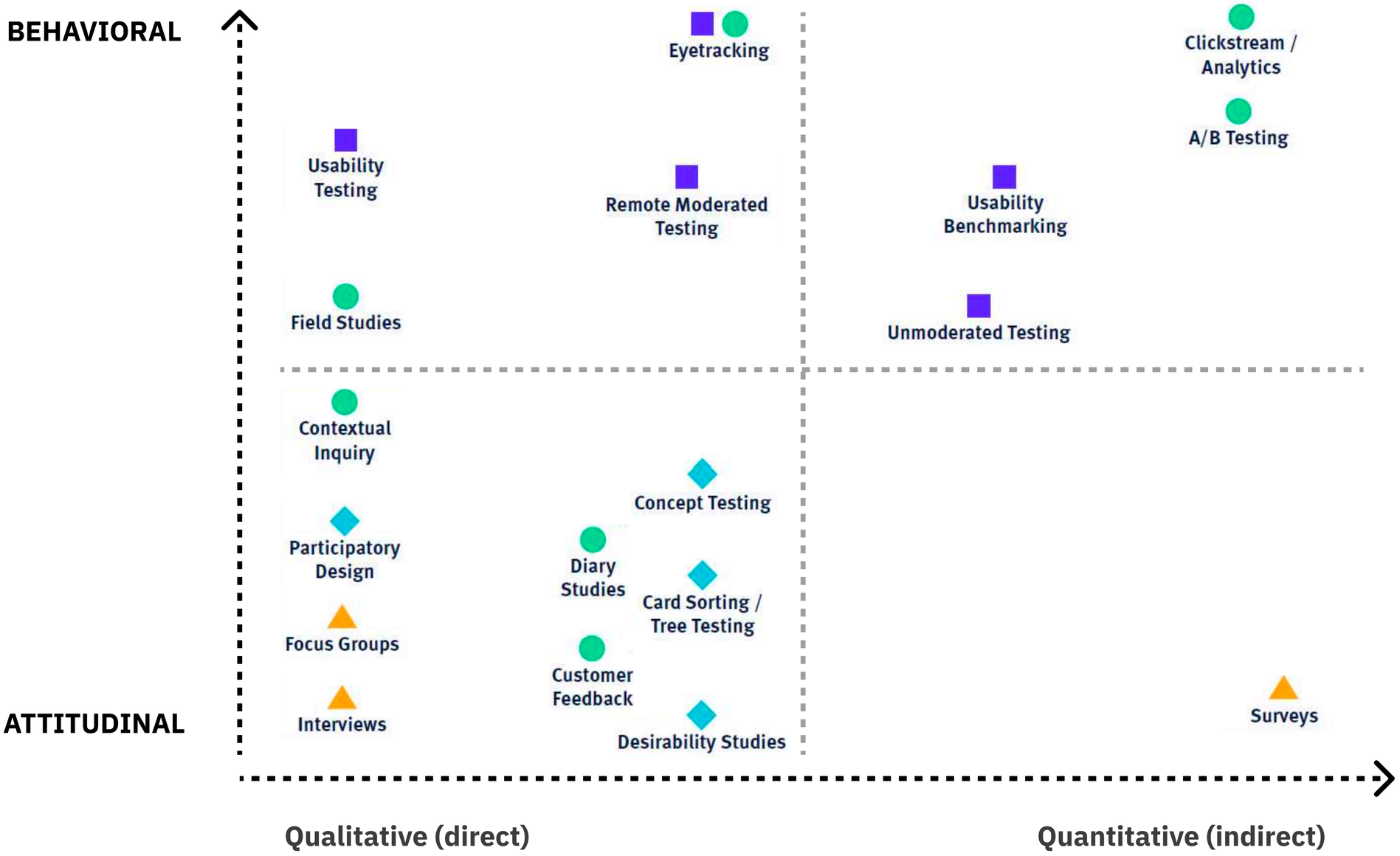
How many & how much

can be counted & measured, expressed using numbers understand the ‘what’



cf. [8]

Popular UX Methods



- Natural use of product
- Scripted (often lab-based) use of product
- Decontextualized / not using product
- Limited use of product

cf. [8]

Some Explanations

Field Studies

“Researchers study participants in their own environment (work or home), where they would most likely encounter the product or service being used in the most realistic or natural environment.” [8]

Contextual Inquiry

“Researchers and participants collaborate together in the participants own environment to inquire about and observe the nature of the tasks and work at hand.” [8]

Diary Studies

“Participants are given a mechanism (diary or camera) to record and describe aspects of their lives that are relevant to a product or service or simply core to the target audience. Diary studies are typically longitudinal and can be done only for data that is easily recorded by participants.” [8]

Interviews (User Interviews)

“A researcher meets with participants one-on-one to discuss in depth what the participant thinks about the topic in question.” [8]

Surveys

“A quantitative measure of attitudes through a series of questions, typically more closed-ended that open-ended.” [8]

Focus Groups

“Groups of 3-12 participants are led through a discussion about a set of topics, giving verbal and written feedback through discussion and exercises.” [8]

Card Sorting

“A quantitative or qualitative method that asks users to organize items into groups and assign categories to each group. This method helps create or refine the information architecture of a site by exposing users’ mental models.” [8]

Tree Testing

“A quantitative method of testing an information architecture to determine how easy it is to find items in the hierarchy. This method can be conducted on an existing information architecture to benchmark it and then again after the information architecture is improved with card sorting to demonstrate improvement.” [8]

Concept Testing

“A researcher shares an approximation of a product or service that captures the key essence (the value proposition) of a new concept or product in order to determine if it meets the needs of the target audience. It can be done one-on-one or with larger numbers of participants, and either in person or online.” [8]

c.f [8];[13]

Research goals

Field Studies
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★ Find new directions & opportunities

▼ Measure product performance

● Improve usability of design

[7];[8]

Research goals

User interviews

Directly ask users about their experiences with a product to understand their thoughts, feelings, and problems

✅ Provides detailed insights that survey may miss
❌ May not represent the wider user base; depends on user's memory and honesty

Focus groups

Gather qualitative data from a group of users discussing their experiences and opinions about a product

✅ Allows for diverse perspectives to be shared and discussed
❌ Group dynamics may influence individual opinions

Field studies

Observe users in their natural environment to inform design decisions with real-world context

✅ Provides contextual insights into user behavior in real-world situations
✅ Helps identify external factors and conditions that influence user experience
❌ Can be time-consuming and resource-intensive to conduct
❌ Participants may behave differently when they know they are being observed (Hawthorne effect)

Diary studies

Get deep insights into user thoughts and feelings by having them keep a product-related diary over a set period of time, typically a couple of weeks

✅ Gives you a peak into how users interact with your product in their day-to-day
❌ Depends on how motivated and dedicated the users are

Research goals

Surveys

Collect quantitative data from a large sample of users about their experiences, preferences, and satisfaction with a product

✅ Provides a broad overview of user opinions and trends
❌ May lack in-depth insights and context behind user responses

Tree testing

Evaluate the findability of existing information within a product's hierarchical structure or navigation

✅ Identifies potential issues in the information architecture
❌ Focuses on navigation structure, not visual design or content

Card sorting

Understand how users categorize and prioritize information within a product or service to structure your information in line with user expectations

✅ Helps create intuitive information architecture and navigation
❌ May not accurately reflect real-world user behavior and decision-making

Concept testing

Evaluate users' reception and understanding of a new product, feature, or design idea before moving on to development

✅ Helps validate and refine concepts based on user feedback
❌ Relies on users' perception and imagination, may not reflect actual use

In short...the most relevant

Check out Examples:
UEX3IL_Deliverable2.pdf on Moodle

recommended

- **User Interview**
  semi-structured,...

...for special use cases

special user group

- **Expert Interviews**
  example: medical field,
  provides in-depth insights
- **Focus Groups**

+ context information

- **Field Study**
  hidden observation,...

recommended

- **Contextual Inquiry**
  observation + interview

re-design

- **Card sorting/**
  Tree testing
  re-design,...

recommended

- **Surveys**
  re-design,...

What qualifies good interview question?

RESEARCH QUESTION

vs.

INTERVIEW QUESTION

RESEARCH QUESTION

“Does drinking coffee affect employee productivity?”

INTERVIEW QUESTION

“Do you like coffee?”

“How many cups of coffee do you drink in a day?”

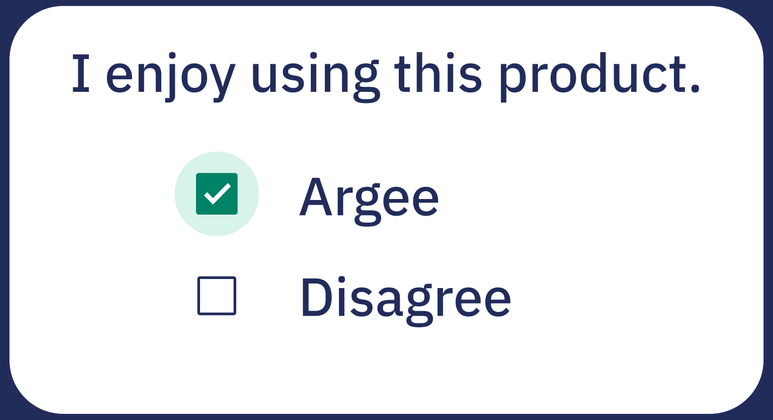
“What’s your favorite brand of coffee?”

survey purposes

Question types

Dichotomous Question

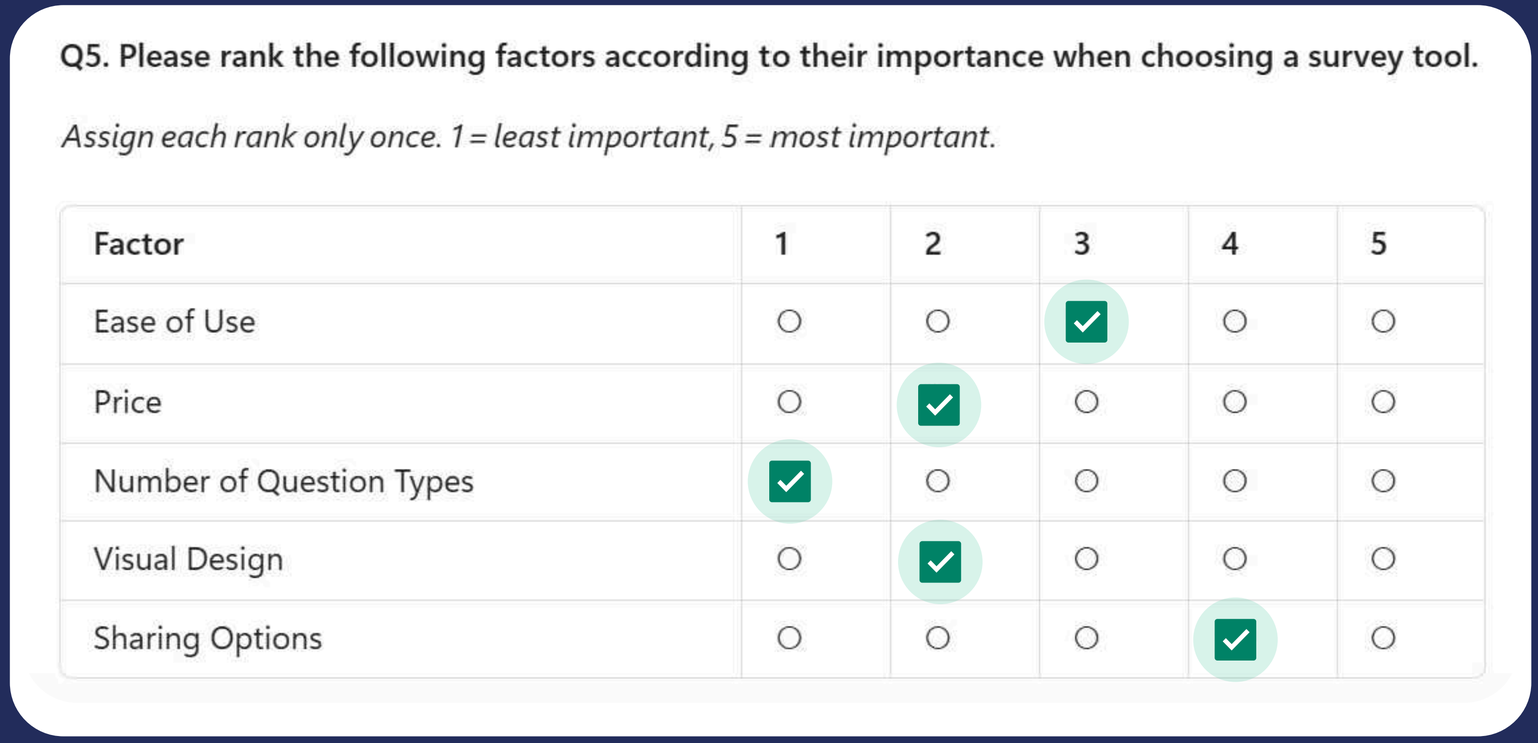
- has only two options (e.g. yes/no, true/false, agree/disagree)
- neutral responses cannot provide
- information limited, but can easily counted (quantitative data)





Ranking Question

- respondents rank a set of given answers based on a defined criteria
- accounts for variations in people’s thoughts and experiences



Q5. Please rank the following factors according to their importance when choosing a survey tool.

Assign each rank only once. 1 = least important, 5 = most important.

Factor	1	2	3	4	5
Ease of Use	○	○	☑	○	○
Price	○	☑	○	○	○
Number of Question Types	☑	○	○	○	○
Visual Design	○	☑	○	○	○
Sharing Options	○	○	○	☑	○

Rating Scale Question

- respondents choose their answer along a scale
- uneven number of answers (mid-point caters to neutral responses)
- special case ‘Likert-Scale’ Questions: 3-/5-/7- point, standardized answers
- accounts for variations in people’s thoughts and experiences

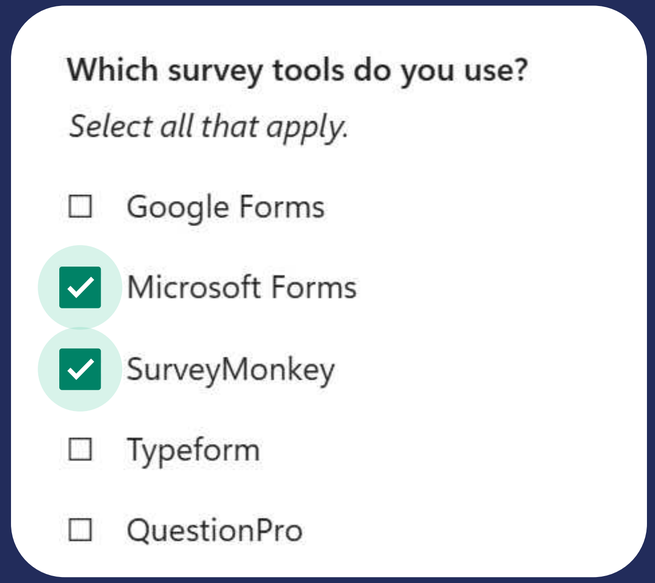




Single- / Multiple- Choice Question

- provides a fixed set of answer options to choose from
- respondents choose one (single-choice) option or multiple options (multiple-choice)
- information limited to answer options, but can easily counted (quantitative data)
- ‘other’ answer option may be combined with text input to get more information





Be aware of different bias

“

3 simple rules to pass the Mom Test:

Talk about their life *instead of your idea*. **Ask about specifics** *in the past instead of generics or opinions about the future.* **Talk less and listen more.**

Rob Fitzpatrick



Failing the Mom Test

“Mom, mom, I have an idea for a business — can I run it by you?”

“Of course, dear.”

“You like your iPad, right? You use it a lot?”

“Yes.”

“Okay, so would you ever buy an app which was like a cookbook for your iPad?”

“Hmmm.”

“And it only costs $40 — that’s cheaper than those hardcovers on your shelf.”

“Well…”

“And you can share recipes with your friends, and there’s an iPhone app which is your shopping list. And videos of that celebrity chef you love.”

[3] p.9

But why?

Failing the Mom Test

“Mom, mom, I have an idea for a business — can I run it by you?”

I am about to expose my ego - please don´t hurt my feeling

“Of course, dear.”

not neutral wording & a big lie

You are my only son and I am ready to lie to protect you

“You like your iPad, right? You use it a lot?”

“Yes.”

You led me to this answer, so here you go

leading/suggestive question

“Okay, so would you ever buy an app which was like a cookbook for your iPad?”

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[3] p.9

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“Yes.”

“Okay, so would you ever buy an app which was like a cookbook for your iPad?”

I am optimistically asking a hypthetical question and you know what I want you to say



hypothetical question

“Hmmm.”

As if I need another cookbook at my age…

not neutral wording & a big lie

“And it only costs $40 — that’s cheaper than those hardcovers on your shelf.”

I´m going to skip that lukewarm signal and tell you more about my great idea



selling idea rather than understanding the person

“Well…”

“And you can share recipes with your friends, and there’s an iPhone app which is your shopping list. And videos of that celebrity chef you love.”

Please just say “yes”. I will not leave you alone until you do



selling idea rather than understanding the person

[3] p.9

Passing the Mom Test

“Hey mom, how’s that new iPad treating you?”

neutral, open-ended question

“Oh - I love it! I use it every day.”

“What do you usually do on it?”

Whoops - we asked a generic question, so answer to this probably won´t be terribly valuable.

generic question

“Oh, you know. Read the news, play sudoku, catch up with my friends. The usual...”

“What’s the last thing you did on it?”

Get specific about examples in the past to get real, concrete data

referring to specific experience in the past

“Did you use an app for that?”

A slightly leading question, but sometimes we need to nudge to get to the topic we´re interested in

“Where did you find out about the other ones you use?”

Dig into interesting and unexpected answers to understand the behaviours and motivations behind

dig depper

[3] p.10

LEADING/SUGGESTIVE

and/or

BIAS QUESTION

Leading/suggestive and/or Bias Questions



Bias in a question means that the question contains a particular viewpoint, assumption or judgement and is therefore NOT entirely NETURAL.

ÄHM..Yes

A **leading question** goes one step further: it actively steers the respondent towards a particular answer or suggests that answer.

Leading/suggestive and/or Bias Questions

Keep in mind:

Every leading question usually contains a bias, but not every biased question is leading.

Leading/suggestive Questions

ÄHM..Yes

The question is posed in such a way that you know exactly what answer is expected. In other words, the way the question is phrased dictates the answer. Leading questions force people to agree with your own opinion.

Characteristics of a leading question:

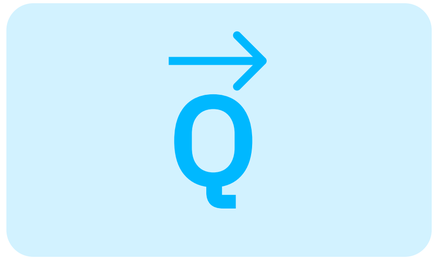
1) Imputation (Questioner thinks he knows the answer)
   a. Beginnings (Do you want, Would you like)
   b. Endings (or?, right? isn´t it?)

2) Reinforcement (a firm opinion)
   a. for sure, certainly

3) Generalisation (many others share this opinion)
   a. ...like everyone else

cf. [1]

Bias Questions



In qualitative research, you should always be aware that your own assumptions can lead to biases that affect data collection. You cannot completely rule out these assumptions – minimise their influence.

cf. [6] own translation



10 minutes Break

[pic 5]

Let´s play a game: The good or bad question Quiz

estimated duration: 5 minutes

mobile-friendly survey

https://forms.cloud.microsoft/e/3JC6jQn9Pf?origin=lprLink

USE



comfortable



broad & focus



simple



open



dig deeper



Mom Test



neutral wording

some Icons by streamline https://home.streamlinehq.com/

AVOID



bias q.



technical terminology



Yes/No a.



hypothetical q.



generic q.



leading/ suggestive q.

some Icons by streamline https://home.streamlinehq.com/

AVOID



guesswork



selling ideas



own ideas



opinions



compliments



fluff: generic claims & promises

some Icons by streamline https://home.streamlinehq.com/

DIG DEEPER



specific events



love & hate



love bad news



facts



frustrations



real-action

some Icons by streamline https://home.streamlinehq.com/

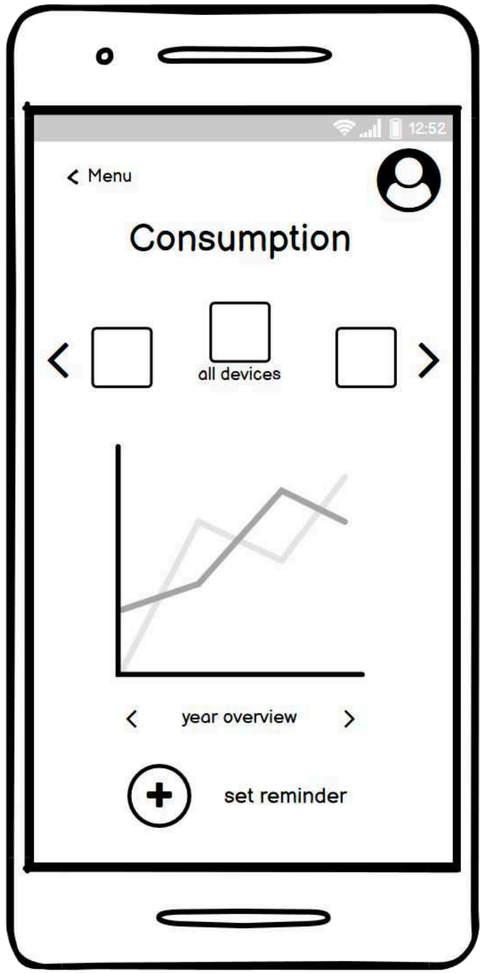
How to Create an Interview questionnaire with AI?


1
PROJECT DESCRIPTION

ENERGY CONTROL APPLICATION

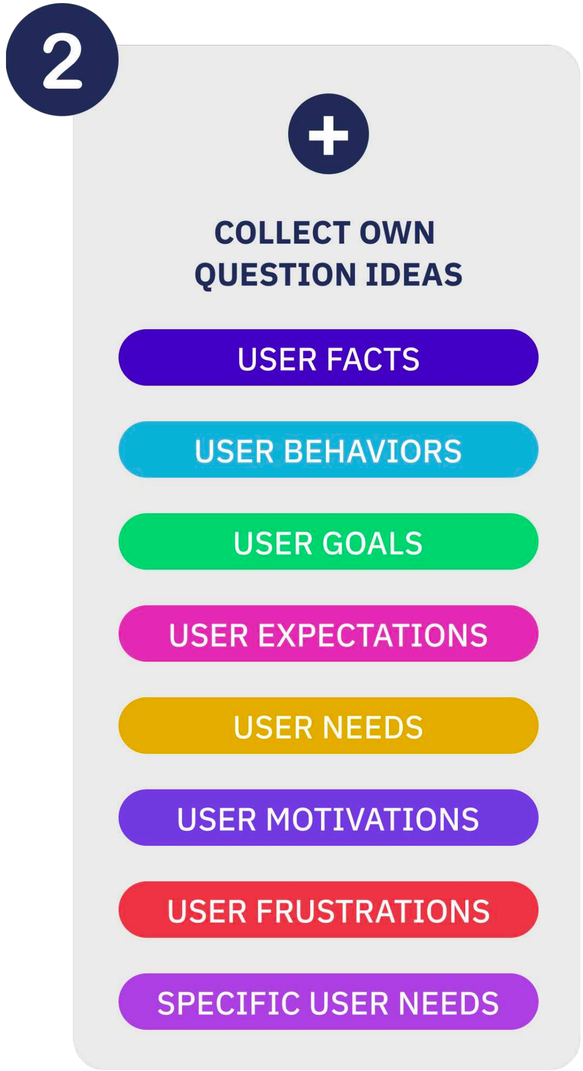
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PROJECT DESCRIPTION



The aim of this project is to develop an Energy Control App that integrates smart home features and enables users to monitor and visualize key energy and water consumption data.

COLLECT OWN QUESTION IDEAS



GENERATE & COMPARE

compare version

3

AI GENERATE QUESTIONNAIRE WITH 2 KI TOOLS

CHOOSE

compare version



GENERATE

compare version

5



FINALIZE USER RESERACH QUESTIONNAIRE

1

PROJECT DESCRIPTION

2

COLLECT OWN QUESTION IDEAS

- USER FACTS
- USER BEHAVIORS
- USER GOALS
- USER EXPECTATIONS
- USER NEEDS
- USER MOTIVATIONS
- USER FRUSTRATIONS
- SPECIFIC USER NEEDS

3

AI GENERATE QUESTIONNAIRE WITH 2 KI TOOLS

4

FILTER

CHOOSE APPROPRIATE QUESTIONS

5

FINALIZE USER RESERACH QUESTIONNAIRE

6

compare version

PILOT TEST

1. **KI-Pilottest** (within 10 fictive test user)
2. **Edge Case KI-Pilottest** (1 fictive a-typical person)
3. **KI-Role Play** (self-check beeing a test user)
4. **Real Pilottest** (UX Design Review test person of another group)

What types of pilot tests are planned for the submission?

INTERNAL PILOT TEST

1. KI-Pilottest (within 10 fictive test user)
2. Edge Case KI-Pilottest (1 fictive a typical person)
3. KI-Role Play (self-check beeing a test user)
4. Real Pilottest (UX Design Review test person of another group)

How to analyze & priorise collected data?

Affinity Diagramming

qualitative data only

The clustering of information, often using sticky notes, into relational groups based on similarities or themes.

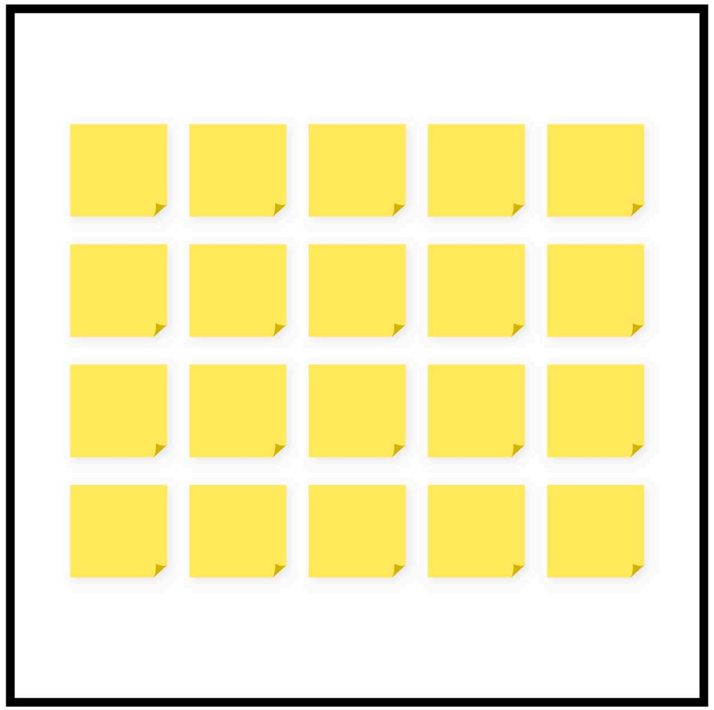


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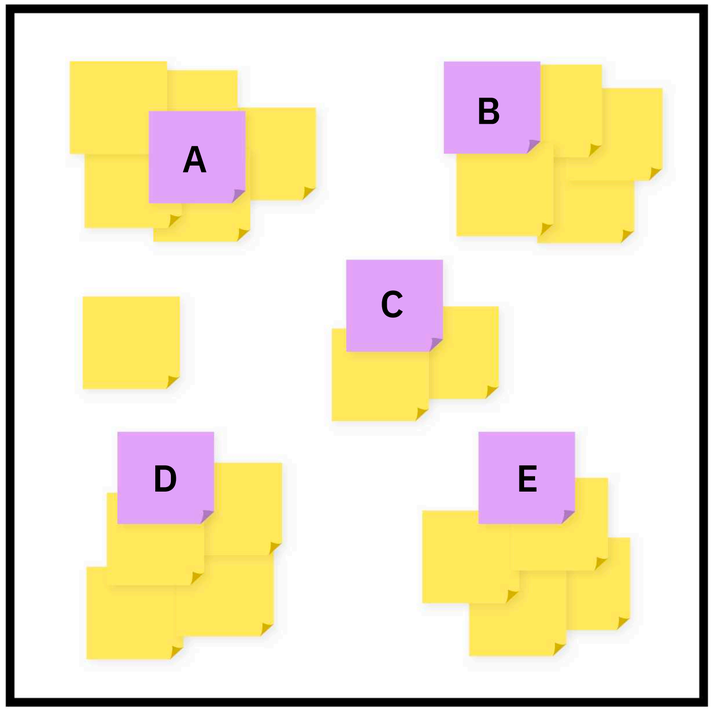
Affinity Diagramming

qualitative data only

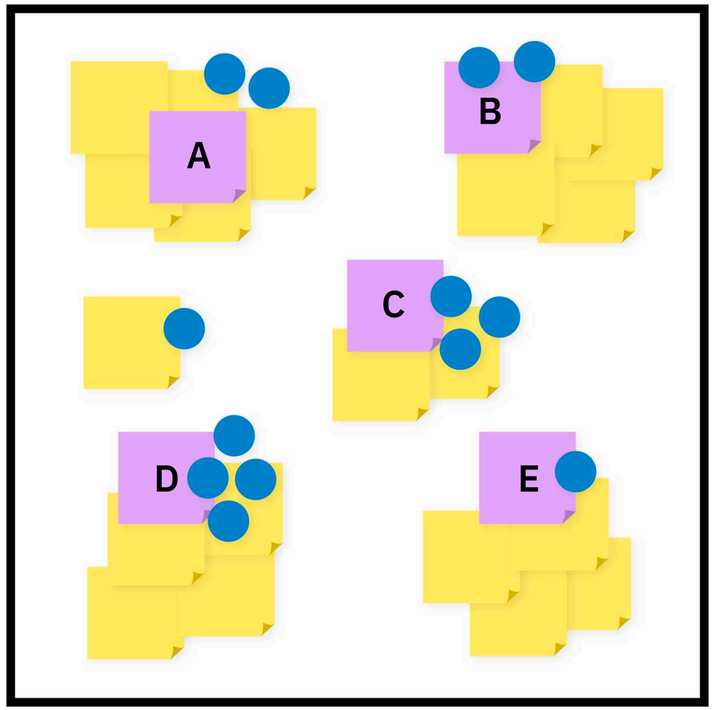
1. Generate Ideas as Sticky Notes



2. Organize Notes into Clusters or Themes



3. Prioritize Clusters and Next Steps



NNGROUP.COM NN/g

[pic 6]

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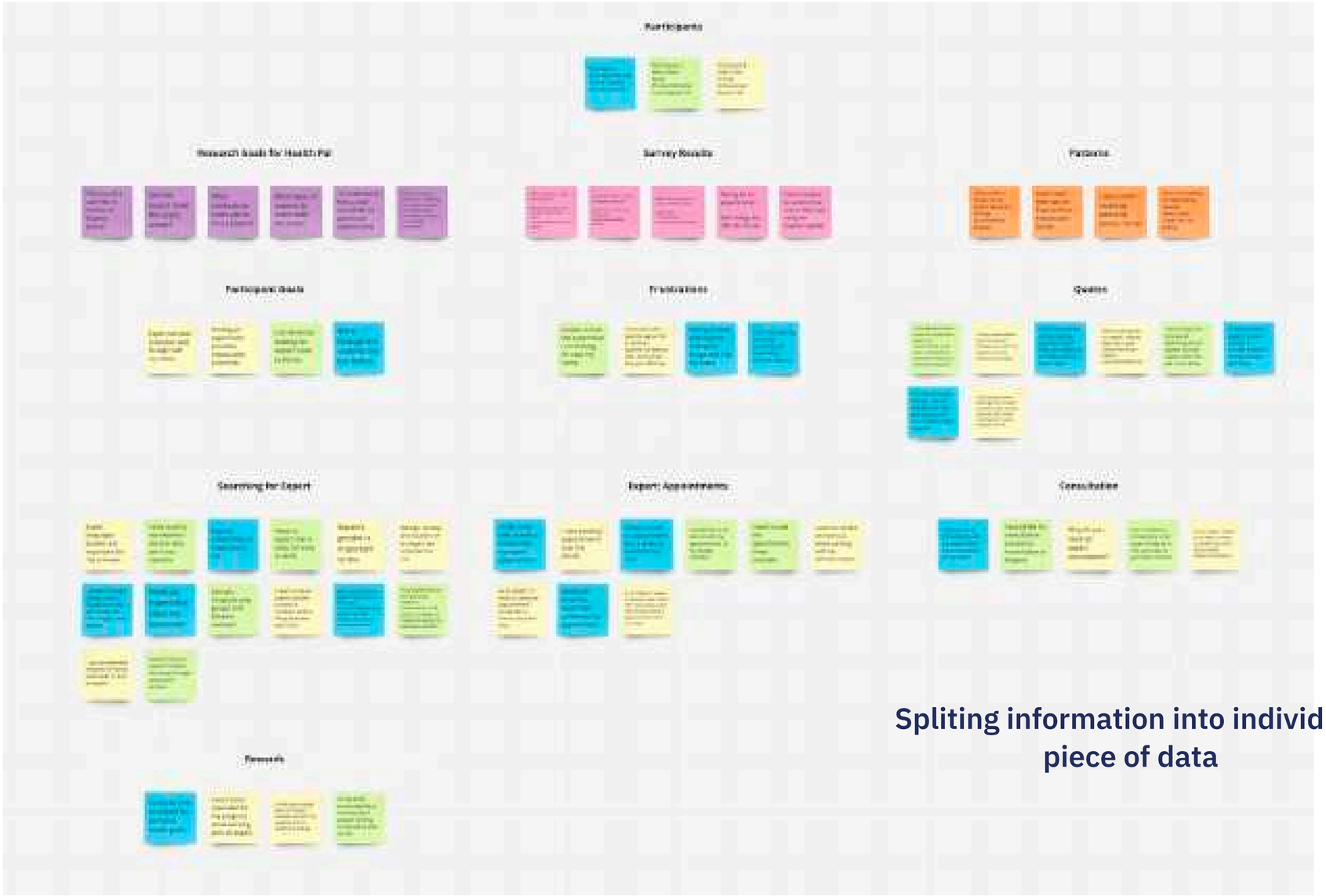


Collaborative aid for decicion making based on research outcomes

[pic 6]

Let´s talk about Best Practices

Best practices



Spliting information into individual piece of data

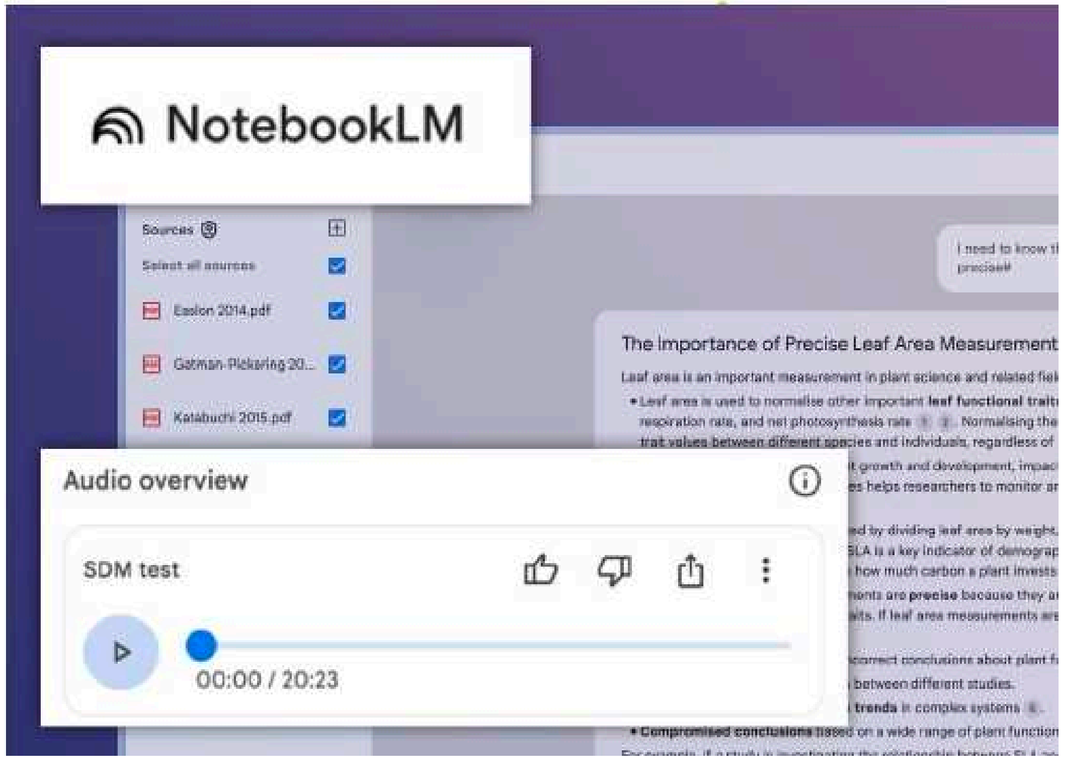
Best practices

Qualitative Analysis Affinity Diagrams

- feeding obtained data into Notebook LM to reveal potential themes + groupings (cross-validate + manually check!)

AI Best Practice

- use Notebook LM as a thinking partner



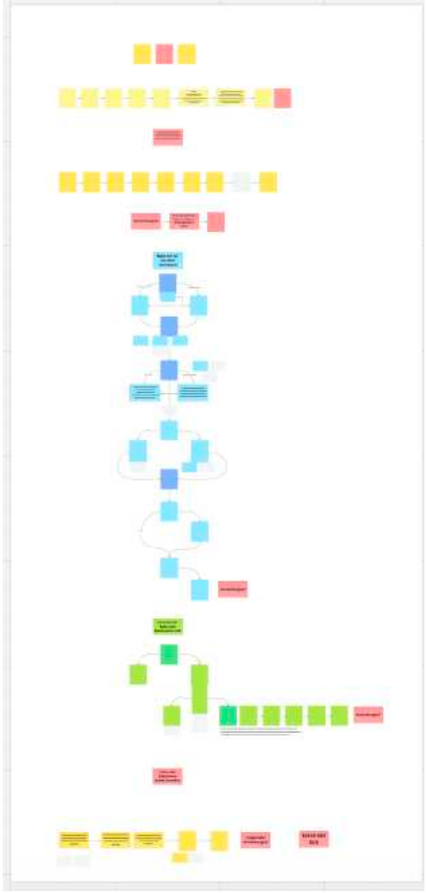
Best practices

User interviews

- structure questionaire to follow a storyline (e.g. introductory, current usage, pro/con, improvement,..) Using a canvas-based tool (e.g. Miro)
- ask yourself for every question: “What can I learn from it?”

AI Best Practice

- generating questions
- manually review quality of questions
- AI for testing questions yourself: have a conversation



Best practices

Survey

- structure survey: Define key points of interest
- ask yourself for every question: “What can I learn from it?”
- What kind of answers do we want? Carefully consider whether to use yes/no and open-ended questions (Question types)
- distribute via appropriate online channels
- limit the amount of questions

AI Best Practice

- generating questions
- manually review quality of questions
- AI for testing questions yourself: have a conversation



Microsoft Forms



Google Forms



SurveyMonkey

UEX3IL 2 - Submission Checklist

- [ ] A) Choose user research method(s)

  (user interview, survey, contextual inquiry...)

- [ ] B) B1. Use two AI tools + compare them + internal Pilottest

  B2. Use one AI tool and iterate your prompts + internal Pilottest

- [ ] C) Conduct your user research

  (2 user interviews of each person in the group)

- [ ] D) Preparation & Analysis

  (question feedback, recording, affinity diagramming...)

References

UEX3IL 2

Literature

[1] Studyflix; Suggestivefrage (Video) Link: https://studyflix.de/jobs/karriere-tipps/suggestivfrage-6174/video

[2] M.Lewrick;P.Link und L.Leifer. The Design Thinking Toolbook. Die besten Werkzeuge & Methoden. ISBN: 978-3-8006-5751-3 Vahlen GmbH Verlag, München 2020.

[3] R.Fitzpatrick. The Mom Test. How to talk to customers and learn if your business is a good idea when everyone is lying to you. (2013) CreateSpace Independent Publishing Platform ISBN/EAN: 9781492180746

[4] D.Travis; P.Hodgson. Think like a UX Researcher. How to observe users, influence design, and shape business strategy. (2019) CRC Press. Taylor & Francis Group ISBN: 978-1-138-36529-2

[6] M.Dempster; H.Donncha. Statistik und Forschungsmethoden für Psychologen und Sozialwissenschaftler für Dummies. (2019) 1.Auflage, WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim ISBN: 978-7-527-71553-4

References

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Literature

[7] C.Rohrer; When to Use Which User-Experience Research Methods. (2022 updated 2026) NNGroup.com URL: https://www.nngroup.com/articles/which-ux-research-methods/

[8] https://media.nngroup.com/media/articles/attachments/User_Research_Methods_A4-compressed.pdf

[9] Maze. What is UX Research: The Ultimate Guide for User Researchers.URL: https://maze.co/guides/ux-research/methods/

[10] NNGroup.com
https://media.nngroup.com/media/articles/attachments/ux_methods_activities_NNg_A4.pdf

[11] S.Farrell; UX Research Cheat Sheet. (2017) NNGroup.com URL: https://www.nngroup.com/articles/ux-research-cheat-sheet/

[12] R.Krause; K. Pernice; Affinity Diagramming for Collaboratively Sorting UX Findings and Design Ideas. (2024) NNGroup.com URL: https://www.nngroup.com/articles/affinity-diagram/

[13] S.Gibbons; Stakeholder Interviews 101. (23 Oct. 2022) NNGroup.com URL: https://www.nngroup.com/articles/stakeholder-interviews/

[14] H.H.Wang; UX Deliverables: Glossary. (09 Aug. 2024) NNGroup.com URL: https://www.nngroup.com/articles/ux-deliverables-glossary/

References

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Pictures

[pic 1] picture KI generated by microsoft copilot chat; [4]p.5
[pic 2] picture KI generated by microsoft copilot chat; [4]p.5
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